

Project 1: Elementary Logic Gates

A typical computer architecture is based on a set of elementary logic gates like And, Or, Mux, etc., as well as on their bitwise versions And16, Or16, Mux16, etc. (assuming a 16-bit machine, like the one built in this course). In this project you will build a typical set of basic logic gates. These gates form the elementary building blocks from which you will build the computer's CPU and RAM chips in later projects.

Below we describe the tools, resources, and implementation tips needed for completing project 1.

Objective

Build the following chips (we use the terms *chip* and *gate* interchangeably):

Nand (given)

Not

And

Or

Xor

Mux

DMux

Not16

And16

Or16

Mux16

Or8Way

Mux4Way16

Mux8Way16

DMux4Way

DMux8Way

Since Nand is considered primitive, there is no need to implement it.

Implementation

Building the chips requires tools, guidelines, and tips. All these resources are described in the [implementation notes](#).